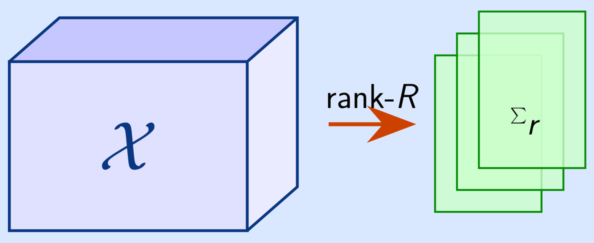
Low-Rank Tensor Approximation & Image Inpainting

Team 4 TP406 2026

1. Introduction & Motivation

- Real-world data arise as **tensors**: RGB images, videos, hyperspectral data
- Low-rank decomposition yields **compact, structured representations**
- This work: CP-ALS and Tucker applied to decomposition & image inpainting



Goal: Given $\mathcal{T} \in \mathbb{C}^{n_1} \otimes \mathbb{C}^{n_2} \otimes \mathbb{C}^{n_3}$,

$$\min_{\hat{\mathcal{T}} \text{ low-rank}} \|\mathcal{T} - \hat{\mathcal{T}}\|_F$$

Contributions:

- CP-ALS decomposition from scratch
- Tucker decomposition (HOSVD / HOOI)
- Mathematical validation & convergence study
- Image inpainting on 6 benchmark datasets

2. CP Decomposition (Canonical Polyadic)

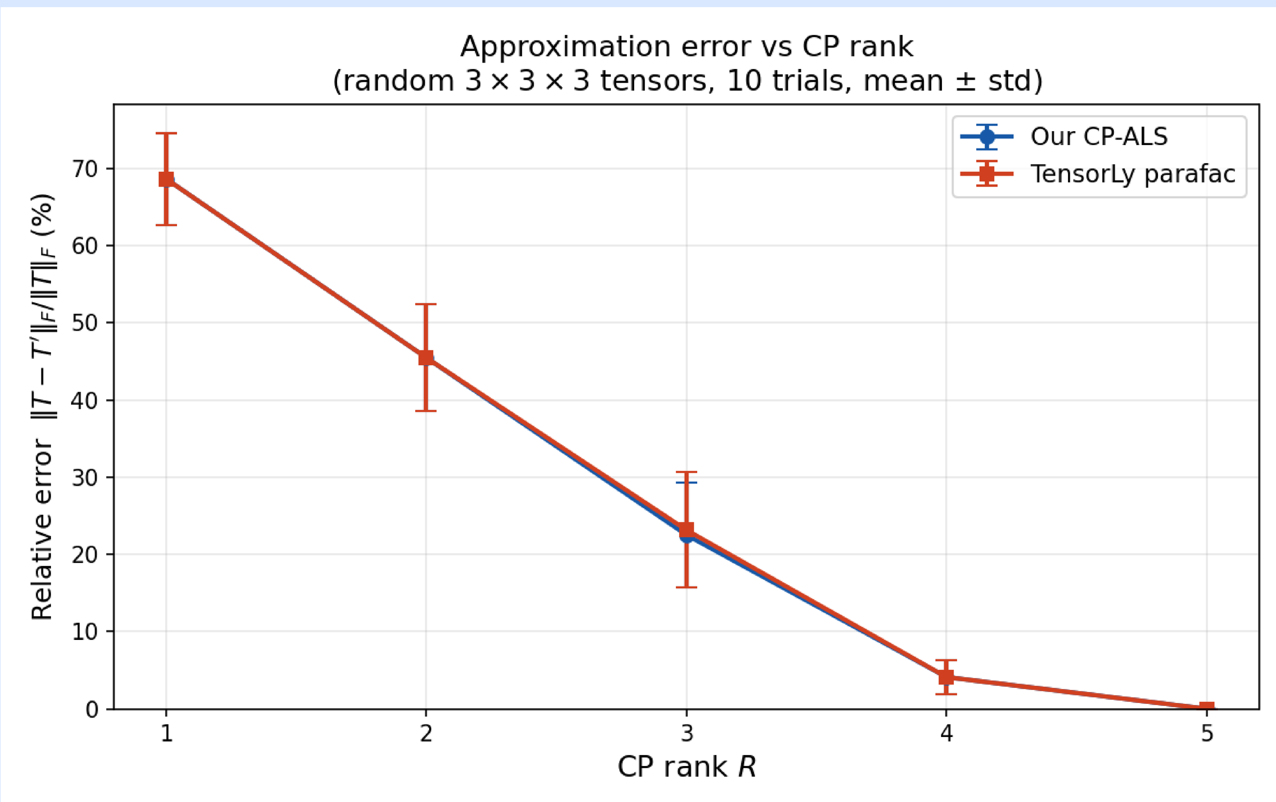
Sum of R rank-one tensors:

$$\mathcal{X} \approx \sum_{r=1}^{R} \mathbf{a}_r \otimes \mathbf{b}_r \otimes \mathbf{c}_r$$

ALS mode-1 update:

$$A \leftarrow X_{(1)}(C \odot B)(B^\top B * C^\top C)^\dagger$$

- Fix $B, C \to$ update A; cycle all modes
- Rank R directly controls compression



CP-ALS matches TensorLy: relative error vs. rank R

3. Tucker Decomposition (HOSVD / HOOI)

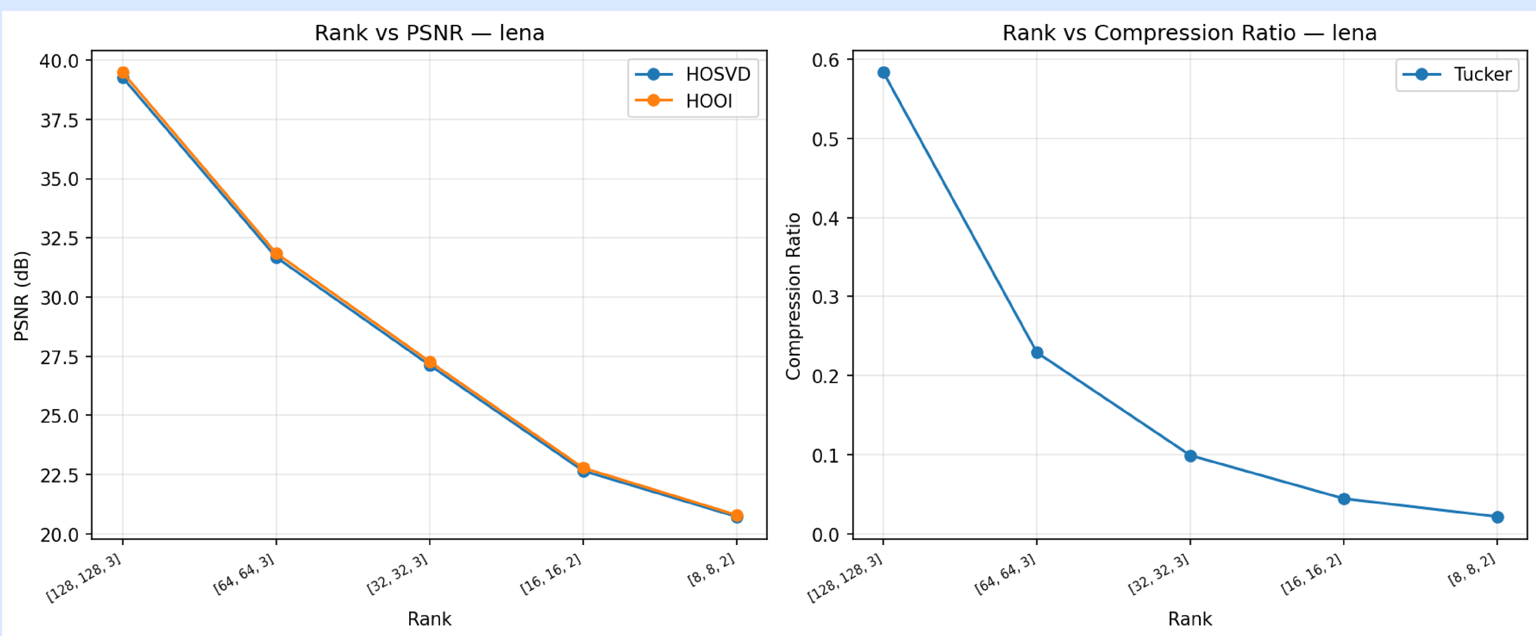
Core tensor $\mathcal{G}$ + factor matrices:

$$\mathcal{X} \approx \mathcal{G} \times_1 A \ \times_2 B \ \times_3 C$$

HOSVD init & HOOI refinement:

$$A^{(n)} = \text{trunc-SVD}_{r_n}(X_{(n)}) \quad \mathcal{G} = \mathcal{X} \times_1 A^\top \times_2 B^\top \times_3 C^\top$$

- Multi-linear rank (r_1, r_2, r_3) — more flexible than CP
- HOSVD: closed-form init; HOOI: iterative refinement



Tucker (Lena): PSNR & compression ratio vs. rank

4. Mathematical Analysis & Validation

Optimization objective:

$$\min_{A,B,C} \left\| \mathcal{X} - \sum_{r=1}^{R} \mathbf{a}_r \otimes \mathbf{b}_r \otimes \mathbf{c}_r \right\|_F^2$$

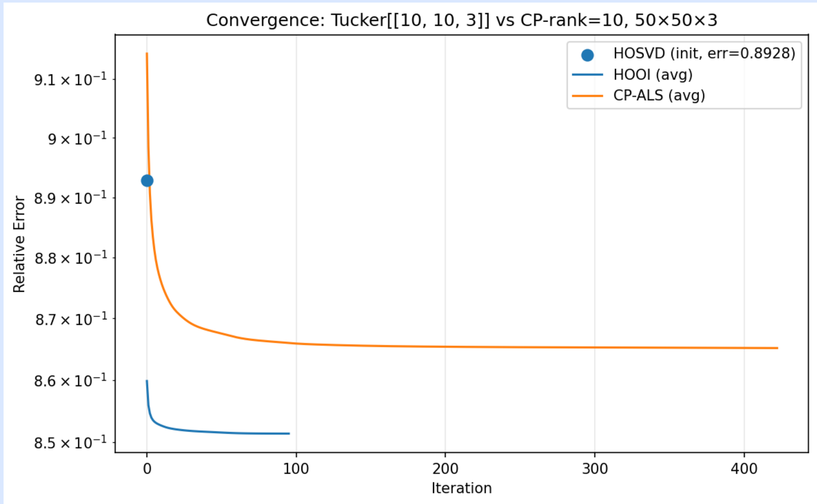
Evaluation metrics:

$$\varepsilon_{\text{rel}} = \frac{\|\mathcal{X} - \hat{\mathcal{X}}\|_F}{\|\mathcal{X}\|_F}$$

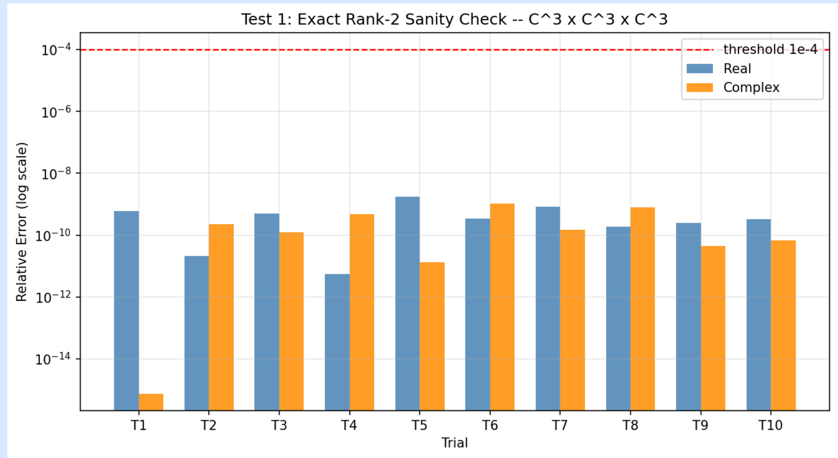
$$\text{PSNR} = 10 \log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}}\right)$$

$$\text{SSIM} = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}$$

ALS iteratively updates each factor matrix while fixing the others, cycling until convergence. Three experiments validate correctness across real and complex tensors: (i) exact rank-2 recovery, (ii) noisy rank-2 recovery, (iii) generic random tensor approximation. All cases satisfy $\varepsilon_{\text{rel}} < 10^{-4}$.



HOOI converges faster than CP-ALS



Exact rank-2: all trials pass $\varepsilon < 10^{-4}$

5. Application: Image Inpainting

Pipeline

- Original Image
- Apply Mask (missing pixels)
- CP-ALS / Tucker Decomposition
- Restored Image

RGB image as 3-D tensor:

$$I \in \mathbb{R}^{H \times W \times 3}$$

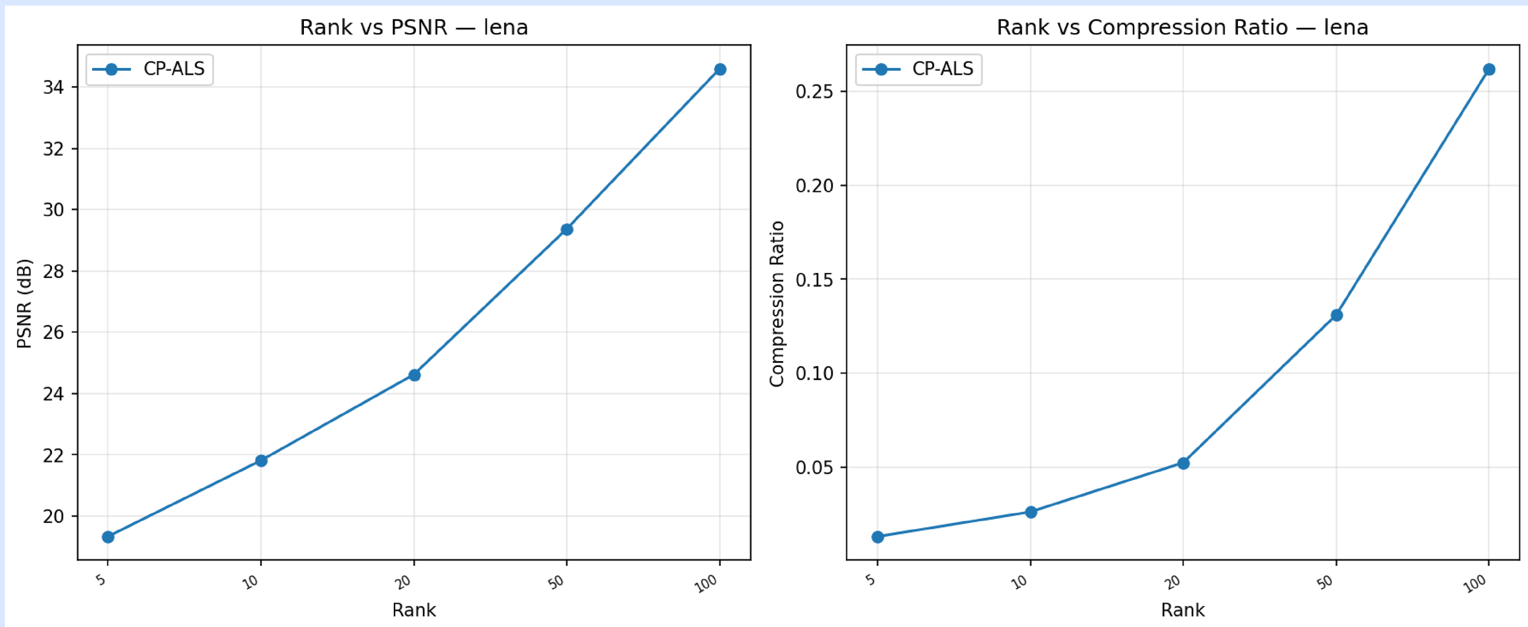
CP-ALS and Tucker decompositions exploit the low-rank structure of RGB image tensors to reconstruct masked (missing) regions. Random masks simulate corruption; performance is quantified by PSNR and SSIM across six standard test images.

Lena: Orig / HOSVD / HOOI / CP Baboon: Orig / HOSVD / HOOI / CP

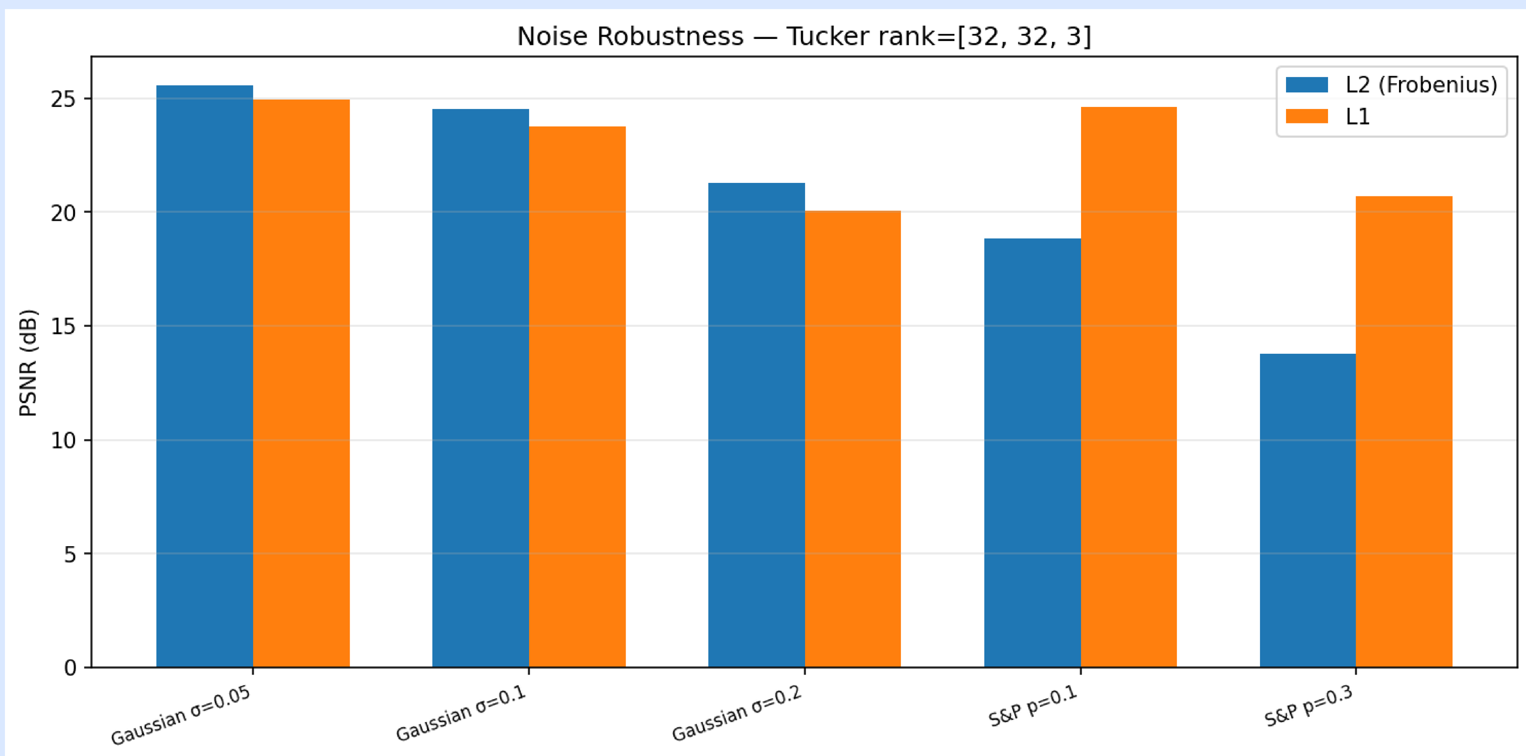
6. Performance Results

Key findings:

- Tucker achieves higher PSNR than CP at the same compression ratio
- L1 (robust) objective significantly outperforms L2 under impulsive Salt-and-Pepper noise
- Rank selection is the dominant hyperparameter controlling the quality–compression trade-off



CP-ALS: PSNR & relative error vs. rank (Lena)



Noise robustness: L1 objective outperforms L2 under S&P noise

7. Conclusion

Tucker decomposition provides greater reconstruction flexibility than CP, and both methods effectively recover missing image pixels via low-rank tensor completion.